

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Third Semester B.Tech (E&C/E&E) (CBCS) Degree Examinations

December 2023

MA2004-1 – VECTOR CALCULUS & TRANSFORM TECHNIQUES

(For E&C, E&E)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- The acceleration at time $t = 1$, of a particle moving along the curve $x = t, y = t$ and $z = -t$ is
 A) $\hat{i} + \hat{j} + \hat{k}$ B) $\vec{0}$
 C) 0 D) $\hat{i} + \hat{j} - \hat{k}$
- The divergence of $\vec{F} = xy^2z\hat{i} + xyz^2\hat{j} + x^2yz\hat{k}$ at $(1, -1, 1)$ is
 A) 1 B) 3
 C) 2 D) 0
- If $\phi = x^2yz$ then $\text{grad}(\phi)$ equals
 A) $2xyz\hat{i} + x^2z\hat{j} + x^2y\hat{k}$ B) $2xyz + x^2z + x^2y$
 C) $2yz\hat{i} + x^2z\hat{j} + x^2y\hat{k}$ D) $yz\hat{i} + x^2z\hat{j} + x^2y\hat{k}$
- If $\vec{F} = yz\hat{i} + xz\hat{j} + xy\hat{k}$ then $\vec{F}$ is
 A) both solenoidal and irrotational B) solenoidal but not irrotational
 C) irrotational but not solenoidal D) neither solenoidal nor irrotational
- If $\vec{F} = (5x^3 - 3y)\hat{i} + 6yz\hat{j} + 8xz^2\hat{k}$ and C is the straight line segment from the origin to the point $(0, 1, 0)$ then $\int_C \vec{F} \cdot d\vec{r}$ is equal to
 A) 0 B) 1
 C) -1 D) 2
- If $\phi = 2x^3yz^2$, $\vec{F} = \nabla\phi$ and C is the circle with centre at the origin and radius 1 then $\int_C \vec{F} \cdot d\vec{r}$ equals
 A) 2 B) 0
 C) 2π D) 1
- If R is a region in the xy -plane, bounded by a simple closed curve C then $\int_C Mdx + Ndy$ equals
 A) $\iint_R \left(\frac{\partial N}{\partial y} - \frac{\partial M}{\partial x} \right) dx dy$ B) $\iint_R \left(\frac{\partial N}{\partial y} + \frac{\partial M}{\partial x} \right) dx dy$
 C) $\iint_R \left(\frac{\partial N}{\partial x} + \frac{\partial M}{\partial y} \right) dx dy$ D) $\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$
- The area of a region R in the xy -plane, bounded by a simple closed curve C is given by
 A) $\frac{1}{2} \int_C y dx + x dy$ B) $\frac{1}{2} \int_C x dy - y dx$
 C) $\frac{1}{2} \int_C y dx - x dy$ D) $\frac{1}{2} \int_C x dx - y dy$
- If $f(z) = u(x, y) + iv(x, y)$ is analytic inside the circle $|z| = 2$ then
 A) $u_x(3, 0) = v_y(3, 0)$ B) $u_x(1, 0) = v_y(1, 0)$
 C) $u_x(5, 0) = v_y(5, 0)$ D) $u_x(0, 3) = v_y(0, 3)$
- If $f(z) = u(x, y) + iv(x, y)$ is an entire function, then
 A) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ B) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$
 C) $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ D) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial x^2} = 0$

11. The bilinear transformation that maps $z = 0, 1, i$ onto $\omega = -1, 0, i$ is
 A) $\omega = \frac{z-1}{z+1}$ B) $\omega = \frac{z+1}{z-1}$
 C) $\omega = \frac{z}{z+1}$ D) $\omega = 1 - \frac{1}{z}$
12. $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)|z|^2$ is equal to
 A) 0 B) 4
 C) -4 D) 1
13. If C is the circle $|z| = 3$ then $\int_C \frac{z^2+2}{z-4} dz$ equals
 A) 0 B) $36\pi i$
 C) $9/\pi i$ D) $22\pi i$
14. If $f(z) = \frac{z^5+5z+2}{(z+1)(z^2-1)}$ then $z = -1$ is a pole of order
 A) 1 B) 2
 C) 3 D) 5
15. If C is the circle $|z| = 3$ then $\int_C \frac{z^3+5}{z^2-1} dz$ equals
 A) 0 B) $2\pi i$
 C) $10\pi i$ D) πi
16. The residue of $f(z) = \frac{z^3+8}{(z-1)(z+2)}$ at $z = 1$ is
 A) 0 B) 9
 C) 3 D) 2
17. If $f(x) = 5x^4$ in $(-\pi, \pi)$ and $f(x+2\pi) = f(x)$ then the coefficient of $\sin 2x$ in the Fourier series expansion of $f(x)$ is
 A) 0 B) $4\pi^4$
 C) $4\pi^5$ D) π^4
18. If u_n and v_n are two sided sequences, then $Z(2u_n + 3v_n)$ equals
 A) $2Z(u_n) + Z(v_n)$ B) $2Z(u_n) + 3Z(v_n)$
 C) $2Z(u_n) - 3Z(v_n)$ D) $3Z(u_n) + 2Z(v_n)$
19. If u_n is a two sided sequence and $Z(u_n) = U(z)$ then $Z(2^{-n}u_n)$ equals
 A) $U(2z)$ B) $U(z/2)$
 C) $U(z)$ D) $U(z+2)$
20. If u_n is a two sided sequence and $Z(u_n) = U(z)$ then $Z(u_{-n})$ equals
 A) $U(-z)$ B) $U(z^{-1})$
 C) $U(z)$ D) $U(z-1)$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I

	Marks	BT*	CO*	PO*
1. a) Prove that the cylindrical coordinate system is orthogonal.	6	L3	1	1
b) Using Stokes theorem evaluate $\int_C (x^2 + y^2) dx - 2xy dy$ where C is the boundary of the rectangle in the xy -plane with vertices $(0, 0), (3, 0), (3, 5)$ and $(0, 5)$.	6	L2	2	2
c) If $\phi = x^2yz^3$ and $\vec{F} = \nabla\phi$ then find $\text{div } \vec{F}$.	4	L2	1	2
2. a) Prove that the vector field $\vec{F} = 2y^3\hat{i} + (6xy^2 + 3z^2)\hat{j} + 6yz\hat{k}$ is conservative. Find its scalar potential and hence find the work done in moving a particle from $A(1, 1, -1)$ to $B(1, 0, 1)$ in the force field $\vec{F}$.	6	L2	2	2
b) State and prove Green's theorem in the plane.	6	L3	2	1
c) Find the angle between the sphere $x^2 + y^2 + z^2 = 9$ and the plane $2x + 3y + 6z = 6$ at the point $(3, 0, 0)$.	4	L2	1	2

3. a) Using Gauss divergence theorem evaluate $\iint_S \vec{F} \cdot \hat{n} \, ds$ where:
 $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ and S is the surface of the cube bounded by the planes $x = 0, x = 1, y = 0, y = 1, z = 0$ and $z = 1$. 6 L2 2 2
- b) Find the work done by the force $\vec{F} = (3x^2y + z)\hat{i} - xz^2\hat{j} + 3xyz\hat{k}$ in moving a particle along the straight line from $(0, 0, 0)$ to $(3, 2, 1)$. 6 L3 2 2
- c) Determine the constants a, b and c if the vector field:
 $\vec{F} = (axy + z^6)\hat{i} + (z - bx^2)\hat{j} + (cy - bxz^5)\hat{k}$ is irrotational. 4 L2 1 2

Unit – II

4. a) Find the bilinear transformation which maps $z = 0, i, -i$ onto $w = 4, -2, 2$. 6 L2 3 2
- b) State and prove Cauchy's integral theorem. 6 L3 4 1
- c) Prove that $u = e^x \cos y$ is a harmonic function. 4 L2 3 2
5. a) Using residue theorem evaluate $\int_C \frac{z^3+5}{(z^2-9)(z^2-1)} dz$ where C is the circle $|z-2| = 2$. 6 L2 4 2
- b) If $f(z) = u(x, y) + v(x, y)$ is analytic in a domain D then prove that at any point in D , $u_x = v_y$ and $u_y = -v_x$. 6 L3 3 1
- c) Determine the analytic function $f(z)$ whose real part is $u = x^2 - y^2 + 6x$. 4 L2 3 2
6. a) Write the Laurent's expansion of $\frac{1}{z^2-3z+2}$ in the region $1 < |z| < 2$. 6 L2 4 2
- b) Find the poles and residues at each pole for the function
 $f(z) = \frac{z^2-6}{(z^2-4)(z-1)^2}$ 6 L2 4 2
- c) Is the function $e^y \cos x + ie^y \sin x$ an entire function? Justify your answer. 4 L2 3 2

Unit – III

7. a) Obtain the Fourier series of $f(x) = \begin{cases} 0 & -1 < x < 0 \\ 1 & 0 < x < 1 \end{cases}$ and $f(x+2) = f(x)$. 6 L2 5 2
- b) Compute a_0, a_1 and a_2 in the Fourier series expansion of the following function.
- | | | | | | | |
|-----|---|----|----|----|----|----|
| x | 0 | 1 | 2 | 3 | 4 | 5 |
| y | 9 | 18 | 24 | 28 | 26 | 20 |
- c) Find the half range Fourier sine series of $f(x) = 2 - x$ in $(0, 2)$. 4 L2 5 2
8. a) Using Z-transform solve $u_{n+2} - 5u_{n+1} + 6u_n = 1, u_0 = 0$ and $u_1 = 1$. 6 L2 5 2
- b) State and prove the convolution theorem for bilateral Z-transform. 6 L3 5 1
- c) Find the Z-transform of $u_n = \begin{cases} 5^{-n} & n \geq 0 \\ 1 & n < 0 \end{cases}$ 4 L2 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
